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B-2

LAB ASSIGNMENT N0. 4

1. Run `nslookup` to obtain the IP address of the web server for the Indian Institute of Technology in Bombay, India: www.iitb.ac.in. What is the IP address of www.iitb.ac.in

2. What is the IP address of the DNS server that provided the answer to your `nslookup` command in question 1 above?

3. Did the answer to your `nslookup` command in question 1 above come from an authoritative or non-authoritative server?

4. Use the `nslookup` command to determine the name of the authoritative name server for the iit.ac.in domain. What is that name? (If there are more than one authoritative servers, what is the name of the first authoritative server returned by `nslookup`)? If you had to find the IP address of that authoritative name server, how would you do so?

```
PS C:\Users\yashj> nslookup www.iitb.ac.in
Server:  UnKnown
Address:  192.168.43.1

Non-authoritative answer:
Name:    www.iitb.ac.in
Addresses:  64:ff9b::6715:7c85
          103.21.124.133

PS C:\Users\yashj> nslookup -type=NS iit.ac.in
Server:  UnKnown
Address:  192.168.43.1

Non-authoritative answer:
iitb.ac.in      nameserver = dns3.iitb.ac.in
iitb.ac.in      nameserver = dns1.iitb.ac.in
iitb.ac.in      nameserver = dns2.iitb.ac.in

dns1.iitb.ac.in internet address = 103.21.125.129
dns2.iitb.ac.in internet address = 103.21.126.129
dns3.iitb.ac.in internet address = 103.21.127.129
PS C:\Users\yashj>
```

5 Locate the first DNS query message resolving the name gaia.cs.umass.edu. What is the packet number[1] in the trace for the DNS query message? Is this query message sent over UDP or TCP?

```
User Datagram Protocol, Src Port: 58030, Dst Port: 53
    Source Port: 58030
    Destination Port: 53
    Length: 45
    Checksum: 0x2818 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 2]
    [Stream Packet Number: 1]
  > [Timestamps]
    UDP payload (37 bytes)
```

Packet number 1
Protocol=UDP

6 Now locate the corresponding DNS response to the initial DNS query. What is the packet number in the trace for the DNS response message? Is this response message received via UDP or TCP?

```
v User Datagram Protocol, Src Port: 53, Dst Port: 58030
    Source Port: 53
    Destination Port: 58030
    Length: 45
    Checksum: 0xa394 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 2]
    [Stream Packet Number: 2]
  > [Timestamps]
    UDP payload (37 bytes)
```

Packet number 2
Protocol=UDP

7 What is the destination port for the DNS query message? What is the source port of the DNS response message?

Destination port (DNS query): 53

Source port (DNS response): 53
53 is the standard DNS port.

8 To what IP address is the DNS query message sent?

```
Internet Protocol Version 4, Src: 10.0.0.44, Dst: 75.75.75.75
    0100 .... = Version: 4
```

9 Examine the DNS query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?

```
    UDP payload (35 bytes)
✓ Domain Name System (query)
    Transaction ID: 0x3c29
  > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  > Queries
    [Response In: 17]
```

10 Examine the DNS response message to the initial query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?

```
Domain Name System (response)
    Transaction ID: 0x3c29
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 0
  > Queries
  > Answers
    [Request In: 15]
    [Time: 23.908000 milliseconds]
```

11 The web page for the base file http://gaia.cs.umass.edu/kurose_ross/ references the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E_2.jpg , which, like the base webpage, is on gaia.cs.umass.edu. What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/? What is the packet number in the trace of the DNS query made to resolve gaia.cs.umass.edu so that this initial HTTP request can be sent to the gaia.cs.umass.edu IP address? What is the packet number in the trace of the received DNS response? What is the packet number in the trace for the HTTP GET request for the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E2.jpg? What is the packet number in the DNS query made to resolve gaia.cs.umass.edu so that this second HTTP request can be sent to the gaia.cs.umass.edu IP address? Discuss how DNS caching affects the answer to this last question.

ANS:- Initial HTTP GET for base file: Packet 22

DNS query for gaia.cs.umass.edu: Packet 15

DNS response for gaia.cs.umass.edu: Packet 17

No.	Time	Source	Destination	Protocol	Length	Info
15	3.325064	10.0.0.44	75.75.75.75	DNS	77	Standard query 0x3c29 A gaia.cs.umass.edu
17	3.348972	75.75.75.75	10.0.0.44	DNS	93	Standard query response 0x3c29 A gaia.cs.umass.edu A 128.119.245.12

HTTP GET for image file: Packet 205

Second DNS query for gaia.cs.umass.edu: None

Current filter: http

No.	Time	Source	Destination	Protocol	Length	Info
22	3.367054	10.0.0.44	128.119.245.12	HTTP	831	GET /kurose_ross/ HTTP/1.1
28	3.395005	128.119.245.12	10.0.0.44	HTTP	857	HTTP/1.1 200 OK (text/html)
205	3.570142	10.0.0.44	128.119.245.12	HTTP	817	GET /kurose_ross/header_graphic_book_8E_2.jpg HTTP/1.1
516	3.670350	128.119.245.12	10.0.0.44	HTTP	454	HTTP/1.1 200 OK (JPEG JFIF image)
520	3.673776	10.0.0.44	128.119.245.12	HTTP	788	GET /favicon.ico HTTP/1.1
524	3.692288	128.119.245.12	10.0.0.44	HTTP	550	HTTP/1.1 404 Not Found (text/html)

Reason: The IP address was already stored in DNS cache, so no second DNS lookup was needed.

12 What is the destination port for the DNS query message? What is the source port of the DNS response message?

```
User Datagram Protocol, Src Port: 57837, Dst Port: 53
    Source Port: 57837
    Destination Port: 53
    Length: 42
    Checksum: 0x0677 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 1]
  > [Timestamps]
    UDP payload (34 bytes)
```

```
User Datagram Protocol, Src Port: 53, Dst Port: 57837
    Source Port: 53
    Destination Port: 57837
    Length: 58
    Checksum: 0x0096 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 2]
  > [Timestamps]
    UDP payload (50 bytes)
```

Destination port (DNS query): 53
Source port (DNS response): 53
53 is the standard DNS port.

13. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

```
Internet Protocol Version 4, Src: 10.0.0.44, Dst: 75.75.75.75
    0100 .... = Version: 4
```

Yes, this is your default local DNS server.

14. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

```
Domain Name System (query)
    Transaction ID: 0x609b
  > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    > www.cs.umass.edu: type A, class IN
    [Response In: 20]
```

15. Examine the DNS response message to the query message. How many "questions" does this DNS response message contain? How many "answers"?

```
v Domain Name System (response)
    Transaction ID: 0x609b
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    > www.cs.umass.edu: type A, class IN
  > Answers
    [Request In: 19]
    [Time: 34.183000 milliseconds]
```

16. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

```
> Internet Protocol Version 4, Src: 10.0.0.44, Dst: 75.75.75.75
> User Datagram Protocol, Src Port: 59963, Dst Port: 53
> Domain Name System (query)
```

Yes, this is the IP address of the default local DNS server.

17. Examine the DNS query message. How many questions does the query have? Does the query message contain any "answers"?

```
Domain Name System (query)
    Transaction ID: 0x6683
  > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  > Queries
    [Response In: 14]
```

18. Examine the DNS response message (in particular the DNS response message that has type "NS"). How many answers does the response have? What information is contained in the answers? How many additional resource records are returned? What additional information is included in these additional resource records (if additional information is returned)?

```
Domain Name System (response)
    Transaction ID: 0x6683
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 3
    Authority RRs: 0
    Additional RRs: 3
  > Queries
  > Answers
  > Additional records
    [Request In: 13]
    [Time: 24.632000 milliseconds]
```

The DNS response contains several answers.
These answers show the names of the authoritative name servers for umass.edu.
Additional records are also included.
These additional records give the IP addresses of those name servers.